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Control Barrier Function

Candidato:
Luca Mombelli
Matricola VR500265

Relatori:
Prof. Nicola Sanosetto

Correlatore:
Mauro Bonaifini

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Capitolo 1

Control Barrier Function

1.1 Nagumo's Theorem

We presented Nagumo's Theorem following [1] , which is one of the fundamental building blocks for control barrier function

Definition 1.1.1. Let K be a subset of a finite dimensional vector-space (or even a normed space) X . We shall say that a function $x(\cdot) : [0, T] \to X$ is *viable* in K on $[0, T]$ if

$$\forall t \in [0, T] \ , \quad x(t) \in K$$

Consider

$$\begin{cases} \dot{x}(t) = f(x(t)) & \forall t \in [0, T] \\ x(0) = x_0 \end{cases} \tag{1.1.1}$$

con $f : \Omega \subset_{\text{op}} X \to X$

Definition 1.1.2. Let K be a subset of Ω. We shall say that K is *locally viable* under f (or enjoys the local viability propery for the map f) if for any initial state $x_0 \in K$, there exists $T > 0$ and a viable solution su $[0, T]$ to the differential equation 1.1.1 with initial condition x_0. It is said to be globally viable under f if we can always take $T = \infty$ The subset K is said to be invariant under f if for any initial state x_0 of K , all solutions to the differential equation 1.1.1 are viable in K

The concept of invariance depends upon the behavior of f on the domain Ω outside K . But we observe that viability property depends only on the behavior of f on K

Definition 1.1.3 (Bouligand Contingent cone)**.** Let X be a normed space , K be a nonempty subset of X and x belong to K . The contingent cone to K at x is the set

$$T_K(x) = \{v \in X \mid \liminf_{h \to 0^+} \frac{d_K(x + hv)}{h} = 0\}$$

where $d_k(x) := \inf_{y \in K} \|x - y\|$

In other worlds , v belongs to $T_K(x)$ if and only if exist a sequence of $h_n > 0$ $h_n \to 0^+$ and a sequence $v_n \in X$, $v_n \to v$ such that

$$\forall n \in \mathbb{N} \ , \ x + h_n v_n \in K$$

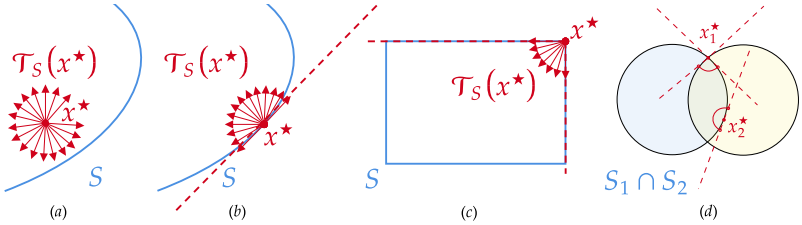


Figura 1.1: Contingent Cone when : (a) $x^\star$ is an interior point , (b) $x^\star$ is on a smooth boundary , (c) $x^\star$ is on a non smooth boundary , (d) for two distinct points $x_1^\star$ and $x_2^\star$ of the intersection $S_1 \cap S_2$

We see that $\forall x \in Int(K), \quad T_K(x) = X$. Therefore , when K is open , the contingent cone to K at any point $xinK$ is always equal to the whole space. Therefore we will be mostly interested in defining the contingent cone for point in the boundary of K . In any case the contingent cone of a point x in the boundary can be the whole space

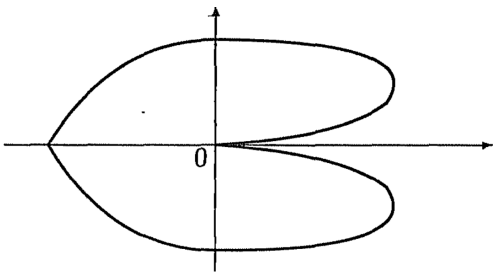


Figura 1.2: Insieme K tale che $T_k(0) = X$

Lemma 1.1.1. Let $x : [0, T] \to K$ be a differentiable viable function

$$\forall t \in [0, T) \quad x'(t) \in T_K(x)$$

Definition 1.1.4 (Viability Domain)**.** Let K be a subset of Ω. We shall say that K is a viability domain of the map $f : \Omega \to X$ se

$$\forall x \in X \quad , \quad f(x) \in T_K(x)$$

Theorem 1.1.1 – Nagumo

Let us assume that

1. K is locally compact
2. f is continuous from K to X

The K is locally viable under f if and only if K is viability domain of f

Esempio 1.1.1. Let $K \subset \mathbb{R}$, with $K = [-2, 2]$, and $\dot{x}(t) = x(t)$. K is a bounded and closed interval, therefore K is compact. Moreover, the function is continuous, therefore the hypotheses of the theorem are satisfied. Let's check whether K is a *viability domain*. We know that if $x \in (-2, 2)$, then $T_K(x) = \mathbb{R}$. Now let's analyze the two boundary points:

- ⋆ If $x = 2 \to f(x) = 2$. The tangent cone represents the directions in which I can move without exiting K. In this case, I can only move to the left, so I obtain $T_K(2) = (-\infty, 0]$.
- ⋆ Following a reasoning similar to the previous one, We obtain $T_K(-2) = [0, +\infty)$.

We can see that for both boundary points we obtain $f(x) \notin T_K(x)$, therefore K is not a viability domain for f.

Now, let's consider the interval $K = (-\infty, 0]$. In this case, the subset is not compact but locally compact, since it is a closed subset of a locally compact space, the Euclidean space. As before, if we take $x \in (-\infty, 0)$, we obtain $T_K(x) = \mathbb{R}$. Let's now move on to study the point $x = 0$, for which we obtain $f(0) = 0$; instead, the tangent cone at the point zero is $T_K(0) = (-\infty, 0]$. We can observe that $f(0) \in T_K(0)$, therefore K is a viability domain, and hence locally viable.

Esempio 1.1.2. Let $K \subset \mathbb{R}^2$ be given by $K = \{(x, y) \in \mathbb{R}^2 \mid x^2 + y^2 - 5 \leq 0\} = \{(x, y) \in \mathbb{R}^2 \mid g(x, y) \leq 0\}$, and let us consider the vector field $f(x, y) = (x, y^2)$. K is a compact subset and f is a continuous vector field.

We know that for interior points $x \in K^\circ$, the tangent cone is $T_K(x) = \mathbb{R}^2$. Let us now consider the boundary points $x \in \partial K$, which belong to the level curve $g^{-1}(0) = \{(x, y) \in \mathbb{R}^2 \mid x^2 + y^2 = 5\}$. In this case, the boundary curve is differentiable, and we can define the tangent cone using tangent lines. Indeed,

$$T_K(x_0) = \{v \in \mathbb{R}^2 \mid g(x_0) + \nabla g(x_0) \cdot v \leq 0\}$$
$$T_K(x_0) = \{v \in \mathbb{R}^2 \mid \nabla g(x_0) \cdot v \leq 0\}$$

Let us compute the condition for the vector field to belong to the tangent cone at the boundary:

$$\begin{aligned} \nabla g(x) \cdot f(x) &= (2x, 2y) \cdot (x, y^2) \\ &= 2x^2 + 2y^3 \leq 0 \\ &\implies y \leq -x^{\frac{2}{3}} \end{aligned}$$

Indeed, evaluating this at the boundary point $x_0 = (0, \sqrt{5})$:

$$\begin{aligned} f(x_0) &= (0, 5) \\ g(x_0) &= 0 \\ \nabla g(x_0) &= (0, 2\sqrt{5}) \\ \nabla g(x_0) \cdot f(x_0) &= 10\sqrt{5} > 0 \implies f(x_0) \notin T_K(x_0) \end{aligned}$$

We have thus shown that K is not a viability domain for f, meaning K cannot be locally viable under f.

Now, consider the same set K but with the vector field $f(x, y) = (-x, -y^3)$. Let us check if this vector field satisfies the condition for belonging to the tangent cone:

$$\begin{aligned} \nabla g(x) \cdot f(x) &= (2x, 2y) \cdot (-x, -y^3)^T \\ &= -2x^2 - 2y^4 \leq 0 \quad \forall (x, y) \in \mathbb{R}^2 \end{aligned}$$

We have therefore shown that $f(x) \in T_K(x)$ for all $x \in K$. Thus, K is a viability domain for f, and by Nagumo's Theorem, K is locally viable under f.

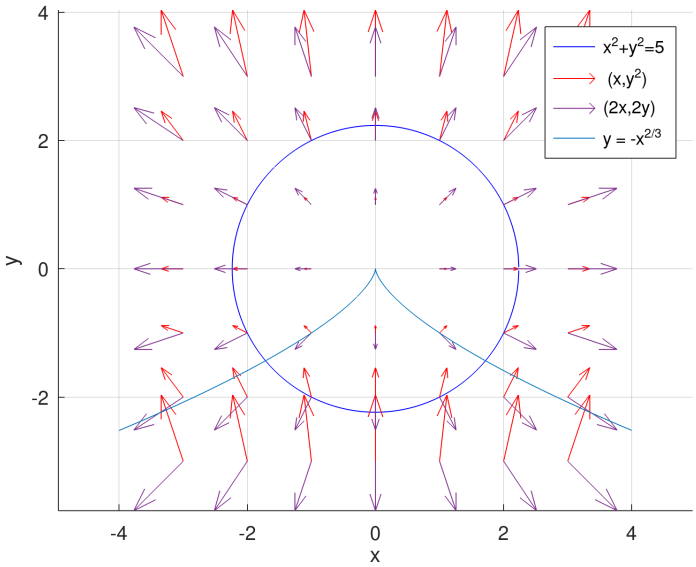


Figura 1.3: Example 1.1.2(a)

Theorem 1.1.2 – Viability

Let us consider a closed subset K of a finite dimensional vector-space X and a continuous map $f : K \to X$.
If K is a viability domain , then for every initial state $x_0 \in K$ there exists a positive T and a viable solution on $[0, T]$ to the differential equation 1.1.1 starting at x_0 such that

$$\begin{cases} T = +\infty \\ T < +\infty \quad \textit{and} \quad \limsup_{t \to T_-} \|x(t)\| = \infty \end{cases}$$

Further adequate information - a priori estimates on the growth of f - allows us to exclude the case when $\lim_{t \to T^-} \|x(t)\| = \infty$.

Theorem 1.1.3

Let us consider a subset of K of a finite dimensional vector-space X and a map f form K to X . We assume that the map f is continuous from K to X , that

$$\exists c > 0 \ t.c \ \forall x \in K \ , \ \|f(x)\| \leq c(\|x\| + 1)$$

and that K is a closed viability domain of f .
Then K is viable under f : for every initial state $x_0 \in K$, there exists a viable solution on $[0, \infty]$ to the differential equation 1.1.1 at x_0

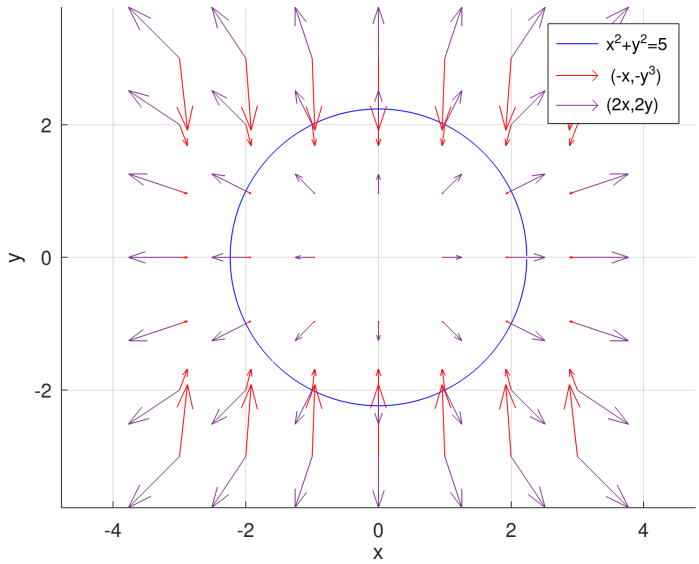


Figura 1.4: Example 1.1.2 (b)

1.2 cbf

Le control barrier function traggono ispirazione dalle control Lyapunov function , che rappresentano l'estensione al controllo delle funzioni di Lyapunov . Allo stesso modo le CBF sono l'estensione al controllo della funzioni di barriera , utilizzate in ottimizzazione. Lo scopo principale delle CBF è definire un sottoinsieme invariante in avanti , ossia un insieme per cui ogni traiettoria che parte da esso rimane al suo interno per ogni tempo futuro . Inoltre sono molto utili nella sintesi di controllori che ci permettono di rimanere nel sottoinsieme.
Consideriamo il sistema dinamico non lineare affine al controllo

Control barrier function are inspired by the control Lyapunov function , which are the extension to control theory of the Lyapunov function. In the same way CBF are the extension to contro theory of the barrier function , used in optimization . The main objective of CBF is to define a forward invariant subset , a subset for which every trajectory that start in the subset is confined in there for all future times . CBF are really useful in the synthesis of controls that let us stay in the foward invariant subset .
Let us consider the non-linear control affine system

$$\dot{x} = f(x) + g(x)u \tag{1.2.1}$$

where $x \in \mathbb{R}^n$, $u \in \mathbb{R}^m$ are respectively the state and the input of the system. The function $f : \mathbb{R}^n \to \mathbb{R}^n$ e $g : \mathbb{R}^n \to \mathbb{R}^{n\times m}$ are locally Lipschitz .
Let us consider the set S , defined as the superlevel set of a continuously differentiable function $h : \mathbb{R}^n \to \mathbb{R}$:

$$S = \{x \in \mathbb{R}^n | h(x) \geq 0\} \tag{1.2.2}$$

$$S^0 = \{x \in \mathbb{R}^n \mid h(x) > 0\} \tag{1.2.3}$$

$$\partial S = \{x \in \mathbb{R}^n \mid h(x) = 0\} \tag{1.2.4}$$

Definition 1.2.1. The set S is forward invariant if for every $x_0 \in S, x(t) \in S$ for $x(0) = x_0$ and all $tinI(x_0)$. The system 1.2.1 is safe with respect to the set S if S is foward invariant

Definition 1.2.2. The $h \in \mathbb{C}^1, h : \mathbb{R}^n \to \mathbb{R}$ is control barrier function , for th system 1.2.1 , on the set S , defined by 1.2.2 - 1.2.4 , if there exists an extended class κ_∞ function α [1] such that :

$$\begin{aligned} &\sup_{u\in\mathbb{R}^m} [L_f h(x) + L_g h(x)u \geq -\alpha(h(x))] \quad \forall x \in S \\ &\sup_{u\in\mathbb{R}^m} [\boldsymbol{\nabla} h(x) \cdot f(x) + \boldsymbol{\nabla} h(x)^T g(x)u \geq -\alpha(h(x))] \quad \forall x \in S \end{aligned} \tag{1.2.5}$$

where

$$L_g h = \begin{bmatrix} L_{g_1} h \\ L_{g_2} h \\ \vdots L_{g_m} h \end{bmatrix}^T$$

so g_i is the i column of g

[1] For $a \in \mathbb{R}^+$, a continuous function $\alpha : [0, a) \to \mathbb{R}$ is a class κ function if it is strictly increasing and $\alpha(0) = 0$. A continuous function $\alpha : (-b, a) \to \mathbb{R}$, with $a, b \in \mathbb{R}^+$, is an extended class κ function if it is strictly increasing and $\alpha(0) = 0$. Furthermore, if $a = b = \infty$ and $\lim_{r\to\infty} \alpha(r) = \infty$, then it is called an extended class κ_∞ function.

Let us consider the set of all the control that render the set S forward invariant :

$$K_{cbf}(u) = \{u \in \mathbb{R}^m \mid L_f h(x) + L_g h(x) u \geq -\alpha(h(x))\}$$

Theorem 1.2.1

Sia $S \subseteq \mathbb{R}^n$ l'insieme di sopralivello di una funzione differenziabile continuità $h : D \subset \mathbb{R}^n \to \mathbb{R}$, se h è una control barrier function su D e $\boldsymbol{\nabla} h(x) \neq 0 \quad \forall x \in \partial S$, allora un qualsiasi controllore liptschiziano u tale che $u(x) \in K_{cbf}$ per il sistema 1.2.1 rende S invariante (in avanti)

Dimostrazione. Per ogni $x \in \partial C$abbiamo che $L_f h(x) + L_g h(x) u \geq -\alpha(h(x)) = 0$. Possiamo allora definire il cono tangente come $T_S(x) = \{v \in \mathbb{R}^n \mid \boldsymbol{\nabla} h(x) v \geq 0\}$. Sappiamo che $\forall x \in S^0 \quad T_S(x) = \mathbb{R}^n$, proviamo a vedere ora i punti appartenti alla frontiera

$$\begin{aligned} \boldsymbol{\nabla} h(x)(f(x) + g(x)u) &\geq 0 \\ \boldsymbol{\nabla} h(x) f(x) + \boldsymbol{\nabla} h(x) g(x) u &\geq 0 \\ L_f h(x) + L_g h(x) u &\geq 0 \end{aligned}$$

allora K è viability domain e quindi locally viable sotto f . Ora abbiamo ipotizzato che f , g , u siano localmente liptschiziane soddisfando così il teorema di esistenza e unicità. Sapendo che la soluzione esista ed è unica le due condizioni di viable e invarianza corrispondo , abbiamo cosi dimostrato ch es è invariante □

Inoltre le control barrier function forniscono una condizione necessaria e sufficiente per l'invarianza in avanti se il sottinsieme S è compatto

Theorem 1.2.2

Sia S un insieme di sopralivello compatto di una funzione $h \in C^1$, $h : \mathbb{R}^n \to \mathbb{R}$, con $\boldsymbol{\nabla} h(x) \neq 0 \quad \forall x \in \partial S$. Se esiste una legge di controllo $u = k(x)$ che rende il sottoinsieme S invariante , allora $h|_S : S \to \mathbb{R}$ è una control barrier function su C

Dimostrazione. Sia $h \in C^1, h|_S : S \to \mathbb{R}$. Per ogni $r \geq 0$ l'insieme $\{x \mid 0 \leq h(x) \leq r\}$ è un sottoinsieme compatto di C. Definiamo una funzione $\alpha : [0, \infty) \to \infty$ con

$$\alpha(r) = - \inf_{\{x \mid 0 \leq h(x) \leq r\}} L_f h + L_g h \; u$$

Usando la compattezza del sottoinsieme e la continuità di $L_f h + L_g h \; u$, α è una funzione ben definita , crescente che soddisfa

$$L_h f + L_g h \; u \geq -\alpha(h(x))$$

Per il Teorema di Nagumo 1.1 l'invarianza di C è equivalente a :

$$L_h f + L_g h \; u \geq 0 \qquad \forall x \in \partial S$$

, ciò implica che $\alpha(0) \leq 0$. Possiamo sempre prendere una funzione di classe κ $\hat{\alpha}$ che maggiora α , facendo così abbiamo ottenuto $L_h f + L_g h \; u \geq -\hat{\alpha}(h(x))$. □

Ora che sappiamo che le CBF forniscono delle condizioni per la sicurezza di un insieme vogliamo trovare un metodo per creare i controlli .
Supponiamo di avere un controllo u per il sistema dinamico 1.2.1 e vogliamo che essere garantisca l'invarianza (sicurezza). Supponiamo che $u \notin K_{cbf}$ per alcune $x \in R^n$. Vogliamo modificare questo controllo per assicurare l'invarianza , iniziamo notando che la condizione 1.2.5 è affine in u , possiamo allora considerare il seguente problema quadratico

$$\begin{aligned} u(x) = \underset{u \in U \subseteq \mathbb{R}^n}{\operatorname{argmin}} \frac{1}{2}\|u - k(x)\|_2^2 \\ s.t \quad [L_f h(x) + L_g h(x) u \geq -\alpha(h(x)) \end{aligned}$$

Per questo problema esiste una soluzione u^* in forma chiusa :

$$u^* = u_{safe} + u_{nom} \tag{1.2.6}$$

dove

$$u_{safe} = \begin{cases} -\frac{\Phi(x,t,u_{des})}{L_g h(x) L_g h(x)^T} L_g h(x)^T & \Phi < 0 \\ 0 \qquad \Phi \geq 0 \end{cases} \tag{1.2.7}$$

con

$$\Phi(x, t, u_{des}) \triangleq \boldsymbol{\nabla} h(x)^T f(x) + \boldsymbol{\nabla} h(x)^T g(x) u + \alpha(h(x)) \tag{1.2.8}$$

We now show an example of the usage of CBF

Esempio 1.2.1. We want to move a material point from an initial position $q_0 \in \mathbb{R}^2$ to a target position $q_g \in \mathbb{R}^2$ while avoiding a spherical obstacle , with center $q_{obs} \in \mathbb{R}^2$ and radius r_{obs} . The system is a single integrator :

$$\dot{q}(t) = u(t)$$

then $f(q) = 0$, $g(q) = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ and $u(t) = \begin{bmatrix} u_1(t) \\ u_2(t) \end{bmatrix}$
We start by introducing the control barrier function

$$h(x) = \|q - q_{obs}\| - r_{obs}$$

which defines the set

$$\begin{aligned} S &= \{q \in R^2 \mid \|q - q_g\| - r_{obs} \geq 0\} \\ &= \{q \in R^2 \mid \sqrt{(x - x_{obs})^2 + (y - y_{obs})^2} \geq r_{obs}\} \end{aligned}$$

We have out nominal control which is

$$u_{nom} = K * \frac{q_g - q}{||q_g - q||} \quad K \in \mathbb{R}_{>0}$$

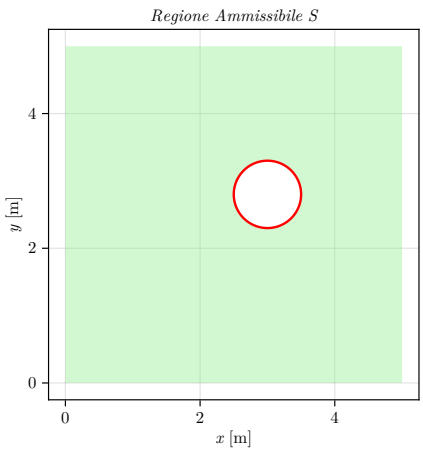


Figura 1.5: The set S

by using equation 1.2.8 and using a constant $\alpha \in \mathbb{R}_{>0}$ as the κ_∞ function we obtain :

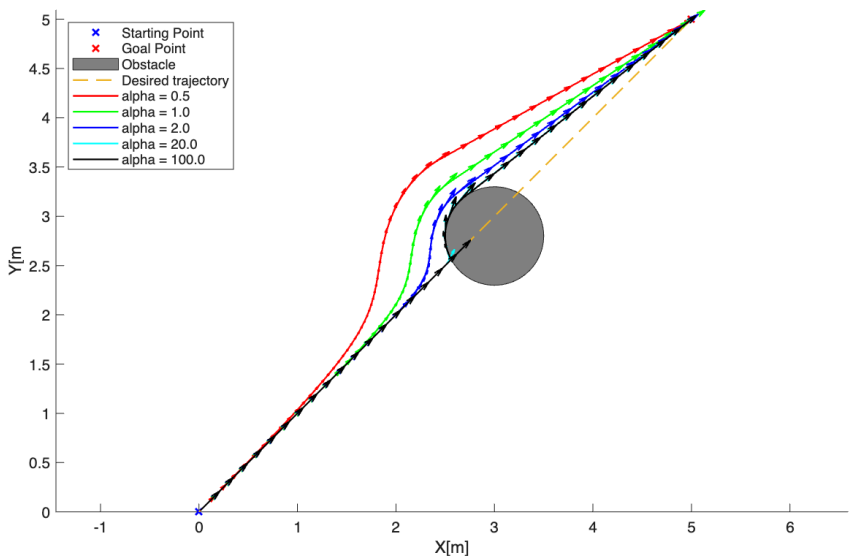
$$\nabla h = \begin{bmatrix} \frac{x}{(\sqrt{x-x_{obs}})^2+(y-y_{obs})^2} \\ \frac{t}{\sqrt{(x-x_{obs})^2+(y-y_{obs})^2}} \end{bmatrix}$$

$$L_g h = \nabla h^T$$

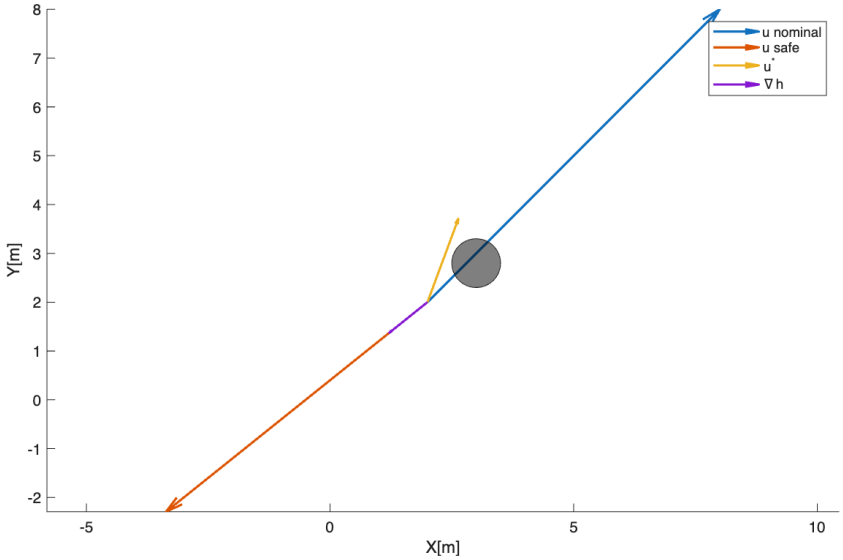
$$L_g h L_g h^T = \frac{x^2+y^2}{(x-x_{obs})^2+(y-y_{obs})^2}$$

$$\Phi = \frac{x}{(\sqrt{x-x_{obs}})^2+(y-y_{obs})^2} u_1 + \frac{y}{(\sqrt{x-x_{obs}})^2+(y-y_{obs})^2} u_2 + \alpha(\|q-q_{obs}\| - r_{obs})$$

and now we have u_{safe} and by adding out u_{nom} we can create our controller that achieves obstacle avoidance while going toward the goal .



In the next figure we will see how every component $(u^*, u_{safe}, u_{nom}, \boldsymbol{\nabla} h)$ is for the point $q = (2, 2)$



1.2.1 Limbo Problem

Equazioni del moto

$$\begin{cases} ml^2\ddot{\theta} - m\ddot{x}l\cos(\theta) - mgl\sin(\theta) = 0 \\ (M+m)\ddot{x} - ml\ddot{\theta}\cos(\theta) + ml\dot{\theta}^2\sin(\theta) = u \end{cases}$$

$$\begin{cases} \ddot{x} = \frac{u + m\sin(\theta)\left(g\cos(\theta) - l\dot{\theta}^2\right)}{M + m\sin^2(\theta)} \\ \ddot{\theta} = \frac{(M+m)g\sin(\theta) - ml\dot{\theta}^2\sin(\theta)\cos(\theta) + u\cos(\theta)}{l\left(M + m\sin^2(\theta)\right)} \end{cases}$$

We find that the equilibrium point are in the form $(x, \theta, u) = (x, 0, 0)$

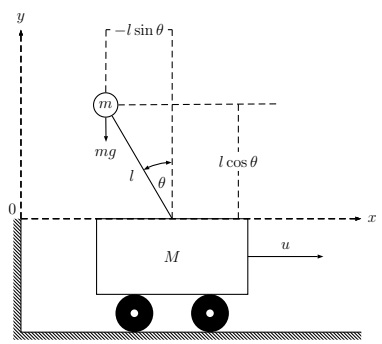


Figura 1.6: Schema del sistema carrello-pendolo inverso.

Angolo Critico

After some calculations we obtain :

$$\ddot{\theta} = \frac{-m\cos(\theta)\sin(\theta)\dot{\theta}^2}{(M+m\sin(\theta)^2} + u\frac{\cos(\theta)}{l(M+m\sin(\theta)^2)} + \frac{(M+m)g\sin(\theta)}{l(M+m\sin(\theta)^2)}$$

We can rewrite by defining the function

$$\begin{aligned} g(\theta) &= \frac{\cos(\theta)}{l(M+m\sin(\theta)^2)} \\ h(\theta,\dot{\theta},u) &= lm\dot{\theta}^2 - \frac{u+g(M+m)\tan(\theta)}{\sin(\theta)} \\ \ddot{\theta} &= -\sin(\theta)g(\theta)h(\theta,\dot{\theta},u) \end{aligned}$$

We want to prove that there exist a crtical anglie $\theta_c \in]-\pi/2,\pi/2[$. We note that $g(\theta) > 0 \quad \forall\theta \in]-\pi/2,\pi/2[$, so if we want the acceleration the have the opposite sign of θ we need to guarantee that $h(\theta,\dot{\theta},u) \geq 0$

$$\begin{aligned} lm\dot{\theta}^2 - \frac{u+g(M+m)\tan(\theta)}{\sin(\theta)} &\geq 0 \\ lm\dot{\theta}^2 &\geq \frac{u+g(M+m)\tan(\theta)}{\sin(\theta)} \\ \text{Now we take the Infimum}(\dot{\theta}=0) \\ \frac{u+g(M+m)\tan(\theta)}{\sin(\theta)} &\leq 0 \end{aligned}$$

We now have two cases

- $\star$ $\sin(\theta) > 0$

$$\begin{aligned} u_{min} \leq u &\leq -g(M+m)\tan(\theta) \\ \tan(\theta) &\leq \frac{-u_{min}}{g(M+m)} \end{aligned}$$

- $\star$ $\sin(\theta) < 0$

$$\begin{aligned} u_{max} \geq u &\geq -g(M+m)\tan(\theta) \\ \tan(\theta) &\geq \frac{-u_{max}}{g(M+m)} \end{aligned}$$

so we obtain

$$\arctan\left(-\frac{u_{max}}{g(M+m)}\right) \leq \theta \leq \arctan\left(-\frac{u_{min}}{g(M+m)}\right)$$

1.3 Geometric Control Theory

Definition 1.3.1. The system 1.2.1 is said to be controllable if for any two points x_0 and x_f in M there exist and admissible controlo $u(t)$ defined on some time interval $[0, T]$ such that the system 1.2.1 with initial condition x_0 reaches the point x_f in time T

Generally it's not easy to prova that a system is contrabble , so we introduce a related property called **accessability** , whisch is often much easier to prove . We first need the notion of reachable set

Definition 1.3.2. Given $x_0 \in M$ we define $R(x_0, t)$ to be the set of all $x \in M$ for which there exists an admissinle controlo u such taht there is a trajectory of 1.2.1 with $x(0) = x_0 \quad , \quad x(t)) = x$. The reachable set form x_0 at time T is definied to be

$$R_T(x_0) = \bigcup_{0 \leq t \leq T} R(x_0, t)$$

Definition 1.3.3. The accessibility algebra C of 1.2.1 is the smallest Lie Algebra of vector field onb M that constains the vector Fields f and $g_1, \ldots, g_m$

The Accessibility algebra is just the span of all possible Lie brackets of f and the g_i

Definition 1.3.4. We define the accessibility distribution C of 1.2.1 to be the distribution generated by the vector field in C , i.e. , $C(x)$ is the span of the vector field X in C at z

Definition 1.3.5. The system 1.2.1 is said to be accessible from $p \in M$ if for evey $T > 0$, $R_T(p)$ contains a non empty open set

Theorem 1.3.1

Consider the system 1.2.1 and assume that the vector field are C^∞. If $\dim C(x_0) = n$ (the accessibility algebra span the tangent space to M at x_0) the for any $T > 0$ the set $R_t(x_0)$ has a nonempry interior ; i.e. , the system ha the accessibility propery from x_0

Definition 1.3.6. A **system** or **machine** $\Sigma = (\mathcal{T}, X, \mathcal{U}, \phi)$ consists of:

- ⋆ A time set $\mathcal{T}$;
- ⋆ A nonempty set X called the **state space** of Σ;
- ⋆ A nonempty set $\mathcal{U}$ called the **control-value** or **input-value space** of Σ; and
- ⋆ A map $\phi : \mathcal{D}_\phi \to X$ called the **transition map** of Σ, which is defined on a subset $\mathcal{D}_\phi$ of

$$\{(\tau, \sigma, x, \omega) \mid \sigma, \tau \in \mathcal{T}, \sigma \leq \tau, x \in X, \omega \in \mathcal{U}^{[\sigma,\tau)}\}, \tag{1.3.1}$$

such that the following properties hold:

nontriviality For each state $x \in X$, there is at least one pair $\sigma < \tau$ in $\mathcal{T}$ and some $\omega \in \mathcal{U}^{[\sigma,\tau)}$ such that ω **is admissible for** x, that is, so that $(\tau, \sigma, x, \omega) \in \mathcal{D}_\phi$;

restriction If $\omega \in \mathcal{U}^{[\sigma,\mu)}$ is admissible for x, then for each $\tau \in [\sigma, \mu)$ the restriction $\omega_1 := \omega|_{[\sigma,\tau)}$ of ω to the subinterval $[\sigma, \tau)$ is also admissible for x and the restriction $\omega_2 := \omega|_{[\tau,\mu)}$ is admissible for $\phi(\tau, \sigma, x, \omega_1)$;

semigroup If σ, τ, μ are any three elements of $\mathcal{T}$ so that $\sigma < \tau < \mu$, if $\omega_1 \in \mathcal{U}^{[\sigma,\tau)}$ and $\omega_2 \in \mathcal{U}^{[\tau,\mu)}$, and if x is a state so that

$$\phi(\tau, \sigma, x, \omega_1) = x_1 \quad \text{and} \quad \phi(\mu, \tau, x_1, \omega_2) = x_2,$$

then $\omega = \omega_1\omega_2$ is also admissible for x and $\phi(\mu, \sigma, x, \omega) = x_2$;

identity For each $\sigma \in \mathcal{T}$ and each $x \in X$, the empty sequence $\diamond \in \mathcal{U}^{[\sigma,\sigma)}$ is admissible for x and $\phi(\sigma, \sigma, x, \diamond) = x$. □

Definition 1.3.7. Let $\mathcal{X}$ be an open subset of $\mathbb{K}^n$ and let $\mathcal{U}$ be a metric space. A **right hand side** with respect to $\mathcal{X}$ is a function

$$f : \mathbb{R} \times \mathcal{X} \times \mathcal{U} \to \mathbb{K}^n$$

which can be obtained in the following way : There must exist another metric space S as well as maps

$$\tilde{f} : S \times \mathcal{X} \times \mathcal{U} \to \mathbb{K}^n$$

and

$$\pi : \mathbb{R} \to S$$

so that

$$f(t, x, u) = \tilde{f}(\pi(t), x, u)$$

and the following properties hold :

- ⋆ $\tilde{f}(s, \cdot, u)$ is of class C^1 for each fixed s,u
- ⋆ both $\tilde{f}$ and the partial derivative $\tilde{f}_x$ are continuos on $S \times \mathcal{X} \times \mathcal{U}$
- ⋆ π is a mueasurable locally essentially bounded function

For f independent of t , the definition reduces to the requirement that U be a metric space and

$$f : \mathcal{X} \times \mathcal{U} \to \mathbb{K}^n$$

and

$$f \quad \text{and} \quad f_x \quad \text{are continuos on} \quad \mathcal{X} \times \mathcal{U}$$

Definition 1.3.8. Let $k = 1, 2, \ldots, \infty$. A **continuos-time system of class** $C^(k)$ is one for which U is an open subset of K^m , for some nonnegative integer m , and for which S and $\tilde{f}$ can be chosen so that S is an open subset of a Euclidean space and $\tilde{f}$ is of class C^k

Proposizione 1.3.1. if Σ is either a continuous-time system , or a discrete system of class C^0 , then Σ is topological when $\mathcal{X}$ is given the induced topologu of $\mathbb{R}^n$

Definition 1.3.9. The system Σ is locally controllable in time T at the equilibrium state x if and only if for every neighborhood $\mathcal{V}$ of x there is another neighboord $\mathcal{W}$ of x , such that , for any pair of elements z , y in W , z can be controlled to y inside $\mathcal{V}$ in time T

Recall the notion of equilibrium state $x \in \mathcal{X}$; for a time-invariant system, this is equivalent to the requirement that there exists a $u \in \mathcal{U}$ such that for all $T \in \mathcal{T}_+$, $\phi(T, 0, x, \omega) = x$, where $\omega \equiv u$. For discrete-time systems this is equivalent to

$$\mathcal{P}(x, u) = x$$

and for continuous-time systems to

$$f(x, u) = 0\,.$$

Theorem 1.3.2

Assume that Σ is a continuos-time system of class C^1 . Then ,

1. Let $\Gamma = (\xi, \omega)$ be a trajectory for Σ on an interval $I = [\sigma, \tau]$. Thean , a sufficient condition for Σ to be locally controllable along ξ is that $\sigma_*[\Gamma]$ be controllable on $[\sigma, \tau]$
2. Ler Σ be time-invariant , and assume that x is an equilibrium state . Pick ant $\varepsilon > 0$ and any $u \in U$ such that $f(x, u) = 0$. Then , a sufficient condition for Σ to be locally controllable at in time ϵ is that the linearization of Σ at (x,u) be controllable

We now define the controllabilty of a linear sisystem

Theorem 1.3.3

The n-dimensionale continuos and invariant time linear system Σ is controllable if and only if $rankR(A, B) = n$ where

$$R(A, B) = [B, AB, A^2B, \ldots, A^{n-1}B]$$

1.3.1 Inverse Pendulum

We denoted with $P = (P_x, P_y) = (x - l\sin(\theta), l\cos(\theta))$ the coordinates of the pendulum and with $d = (d_x, d_y)$ the coordinates of the center of the obstacle of radius ϵ . The state vector is $z = [x, \theta, \dot{x}, \dot{\theta}]$

$$\begin{aligned}
h_0(z) &= (x - l\sin(\theta) - d_x)^2 + (l\cos(\theta) - d_y)^2 - \epsilon^2 \\
&= x^2 + l^2\sin^2(\theta) - d_x^2 - 2xl\sin(\theta) - 2xd_x + 2d_xl\sin(\theta) + \\
&+ l^2\cos^2(\theta) + d_y^2 - 2d_yl\cos(\theta) - \epsilon^2 \\
&= x^2 + l^2 + d_x^2 + d_y^2 - 2(xl\sin(\theta) + xd_x - d_xl\sin(\theta) + d_yl\cos(\theta)) - \epsilon^2
\end{aligned}$$

We compute

$$\nabla h_0(z) = \begin{bmatrix} 2x - 2lsin(\theta) - 2d_x \\ -2(xl\cos(\theta) - d_xl\cos(\theta) - dyl\sin(\theta)) \\ 0 \\ 0 \end{bmatrix}$$

Then we need to compute the term

$$\nabla h_0 \cdot (f + gu) =$$
$$= \begin{bmatrix} 2x - 2lsin(\theta) - 2d_x \\ -2(xl\cos(\theta) - d_x l\cos(\theta) - dyl\sin(\theta)) \\ 0 \\ 0 \end{bmatrix} \cdot \left(\begin{bmatrix} v_x \\ v_\theta \\ \frac{m\sin(\theta)(g\cos(\theta) - lv_\theta^2)}{M + m\sin^2(\theta)} \\ \frac{(M+m)g\sin(\theta) - mlv_\theta^2\sin(\theta)\cos(\theta)}{l(M + m\sin^2(\theta))} \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ \frac{1}{M + m\sin^2(\theta)} \\ \frac{\cos(\theta)}{l(M + m\sin^2(\theta))} \end{bmatrix} u \right)$$
$$= v_x(2x - 2lsin(\theta) - 2d_x) - 2v_\theta(xl\cos(\theta) - d_x l\cos(\theta) - dyl\sin(\theta))$$

and we obtain

$$\begin{aligned} h_1 &= \dot{h_0} + k_1 h_0 \quad k_1 \in \mathbb{R} \\ &= \nabla h_0 \cdot (f + gu) + k_1 h_0 \\ &= v_x(2x - 2lsin(\theta) - 2d_x) - 2v_\theta(xl\cos(\theta) - d_x l\cos(\theta) - dyl\sin(\theta)) + \\ &+ k_1(x^2 + l^2 + d_x^2 + d_y^2 - 2(xl\sin(\theta) + xd_x - d_x l\sin(\theta) + d_y l\cos(\theta))) - \epsilon^2 \end{aligned}$$

Now we define the safe set as $S = S_1 \cap S_2$ where

$$\begin{aligned} S_1 &= \{z \in R^4 \mid h_0(z) \geq 0\} \\ &= \{z \in R^4 \mid x^2 + l^2 + d_x^2 + d_y^2 - 2(xl\sin(\theta) + xd_x - d_x l\sin(\theta) + d_y l\cos(\theta)) \geq \epsilon^2\} \end{aligned}$$

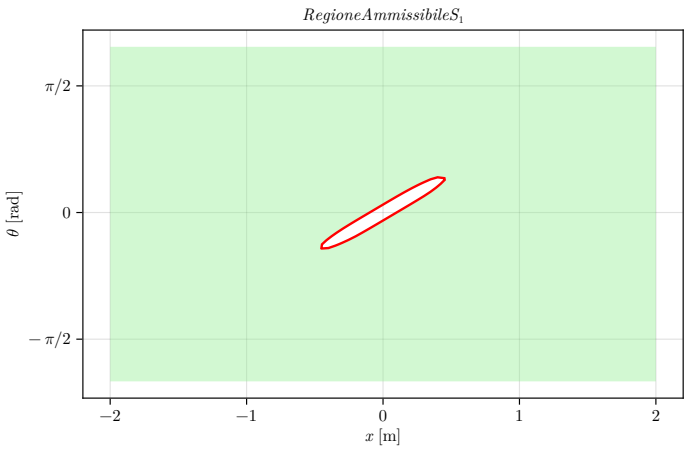


Figura 1.7: The safe set S_1 colored in green

$$\begin{aligned} S_2 &= \{z \in R^4 \mid h_1(z) \geq 0\} \\ &= \{z \in R^4 \mid v_x(2x - 2lsin(\theta) - 2d_x) - 2v_\theta(xl\cos(\theta) - d_x l\cos(\theta) - dyl\sin(\theta)) + \\ &+ k_1(x^2 + l^2 + d_x^2 + d_y^2 - 2(xl\sin(\theta) + xd_x - d_x l\sin(\theta) + d_y l\cos(\theta))) \geq \epsilon^2\} \end{aligned}$$

We also have definied another control barrier function $b_0(z) = \theta_{cri}^2 - \theta^2$. We need to compute the gradient and the term $dot(b_0)$in this case

$$\nabla b_0 = \begin{bmatrix} 0 \\ -2\theta \\ 0 \\ 0 \end{bmatrix}$$

$$\begin{aligned} \nabla b_0 \cdot (f + gu) = \\ &= \begin{bmatrix} 0 \\ -2\theta \\ 0 \\ 0 \end{bmatrix} \cdot \left(\begin{bmatrix} v_x \\ v_\theta \\ \frac{m\sin(\theta)(g\cos(\theta) - lv_\theta^2)}{M + m\sin^2(\theta)} \\ \frac{(M+m)g\sin(\theta) - mlv_\theta^2\sin(\theta)\cos(\theta)}{l(M + m\sin^2(\theta))} \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ \frac{1}{M + m\sin^2(\theta)} \\ \frac{\cos(\theta)}{l(M + m\sin^2(\theta))} \end{bmatrix} u \right) \\ &= -2v_\theta\theta \end{aligned}$$

Then we can define

$$b_1 = -2v_\theta\theta + alpha_1(\theta_{cri}^2 - \theta^2)$$

The safe set in this caso is given by $B = B_1 \cap B_2$

$$B_1 = \{z \in \mathbb{R}^4 \mid \theta_{cri}^2 \geq \theta^2\}$$

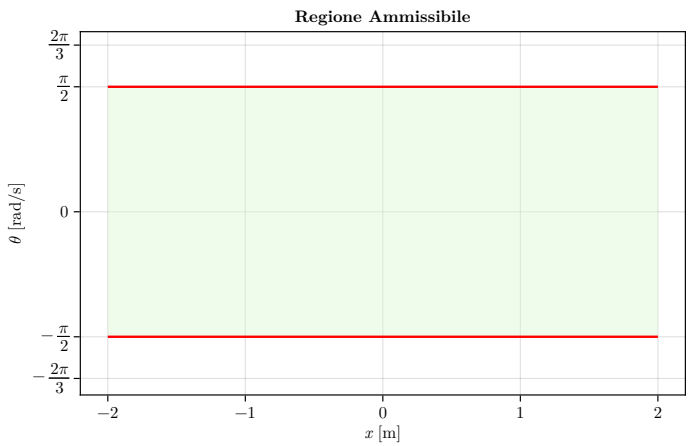


Figura 1.8: The set B_1 colored in green

$$B_2 = \{z \in \mathbb{R}^4 \mid -2v_\theta\theta + \alpha_1(\theta_{cri}^2 - \theta^2) \geq 0\}$$

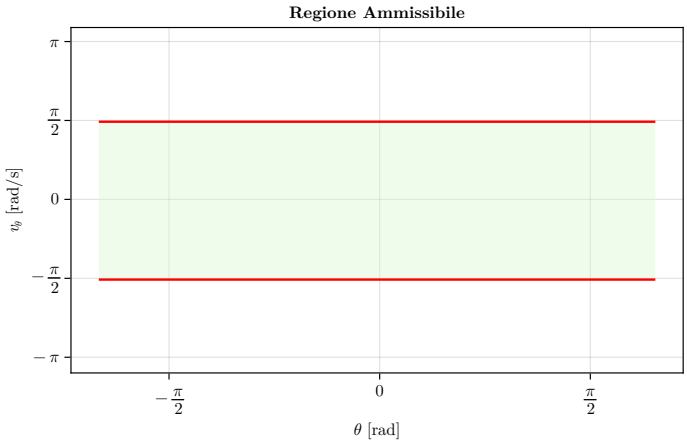


Figura 1.9: The set B_2 colored in green with $\alpha_1 = 40$

The Final boss is $C = S \cap B$ which i refused to write

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